From the definition of $y = \sin^{-1} x$, we observe the following:

(i) $y = \sin^{-1} x$ if and only if $x = \sin y$ for $-1 \le x \le 1$ and $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$.

(ii) $\sin\left(\sin^{-1} x\right) = x$ if $|x| \le 1$ and has no sense if $|x| > 1$.

(iii) $\sin^{-1}\left(\sin x\right) = x$ if $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$. Note that $\sin^{-1}(\sin 2\pi) = 0 \neq 2\pi$.

(iv) $\sin^{-1}\left(\sin x\right) = \pi - x$ if $\frac{\pi}{2} \le x \le \frac{3\pi}{2}$. Note that $-\frac{\pi}{2} \le \pi - x \le \frac{\pi}{2}$.

(v) $y = \sin^{-1} x$ is an odd function.

Remark

Let us distinguish between the equations $\sin x = \frac{1}{2}$ and $x = \sin^{-1}\left(\frac{1}{2}\right)$. To solve the equation $\sin x = \frac{1}{2}$, one has to find all values of x in the interval $(-\infty, \infty)$ such that $\sin x = \frac{1}{2}$. However, to find x in $x = \sin^{-1}\left(\frac{1}{2}\right)$, one has to find the unique value x in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $\sin x = \frac{1}{2}$.

4.3.4 Graph of the inverse sine function

The inverse sine function, $\sin^{-1} : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, receives a real number x in the interval $[-1, 1]$ as input and gives a real number y in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ as output. As usual, let us find some points (x, y) using the equation $y = \sin^{-1} x$ and plot them in the xy-plane. Observe that the value of y increases from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ as x increases from -1 to 1. By connecting these points by a smooth curve, we get the graph of $y = \sin^{-1} x$ as shown in Fig. 4.6.

x	y
-1	$-\frac{\pi}{2}$
$-\frac{\sqrt{2}}{2}$	$-\frac{\pi}{4}$
0	0
$\frac{\sqrt{2}}{2}$	$\frac{\pi}{4}$
1	$\frac{\pi}{2}$

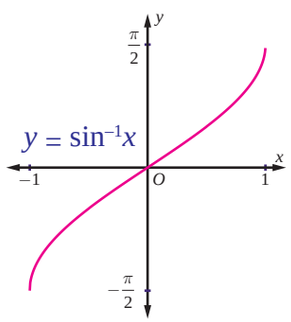


Fig. 4.6

Note

The graph of $y = \sin^{-1} x$

(i) is also obtained by reflecting the portion of the entire graph of $y = \sin x$ in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ about the line $y = x$ or by interchanging x and y axes from the graph of $y = \sin x$.

(ii) passes through the origin.

(iii) is symmetric with respect to the origin and hence, $y = \sin^{-1} x$ is an odd function.

12th_Maths_EM_Vol1_CH 4 Inverse Trigonometry.indd 135 16-12-2022 13:11:59